Sets 1 — Set notation · Venn diagrams

Notation: $\in$ is a member of · $\notin$ is not a member of · $n(A)$ number of elements · $\varnothing$ empty set · $\mathscr{E}$ universal set · A' complement · $A \cap B$ intersection · $A \cup B$ union · $B \subset A$ subset. Marks in brackets. ★ = harder question.

1 Say whether each statement is true or false. [4]

a $7 \in \{\text{prime numbers}\}$

b $15 \notin \{\text{multiples of } 3\}$

c square $\in \{\text{quadrilaterals}\}$

d $0.5 \in \{\text{integers}\}$

> **a** True. **b** False — $15 = 3 \times 5$ is a multiple of 3. **c** True — a square has four sides. **d** False — integers are whole numbers.

2 $\mathscr{E} = \{\text{integers from 1 to 12}\}$, $A = \{\text{even numbers}\}$, $B = \{\text{multiples of } 3\}$. [6]

a List the set $A \cap B$.

b List the set $A \cup B$.

c Find $n(B')$.

d Describe the set $A \cap B$ in words.

> $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$.
> **a** In both: $A \cap B = \{6, 12\}$. **b** In either: $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$. **c** B' = everything not a multiple of 3: $12 - 4 = 8$ elements. **d** The multiples of 6 (even *and* a multiple of 3).

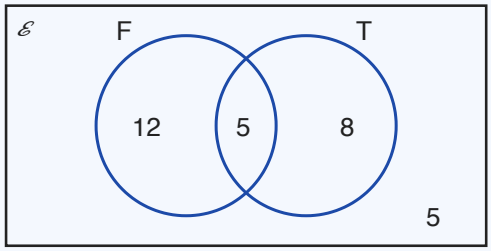
3 In a class of 30 students, $F = \{\text{plays football}\}$ and $T = \{\text{plays tennis}\}$. 12 students play football only, 5 play both, and 8 play tennis only. [5]

a Draw a Venn diagram showing this information.

b How many students play neither sport?

c Find $n(F \cup T)$.

d Find $n(T)$.

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> **b** $30 - (12 + 5 + 8) = 5$ students play neither.
> **c** $n(F \cup T) = 12 + 5 + 8 = 25$. **d** $n(T) = 5 + 8 = 13$ (everyone inside the T circle).

4 $\mathscr{E} = \{\text{cars in a car park}\}$, $R = \{\text{red cars}\}$, $E = \{\text{electric cars}\}$. Describe in words: [4]

- **a** the set R'
- **b** the set $R \cap E$
- **c** the set $R \cup E$
- **d** what it means if $R \cap E = \varnothing$.

> **a** The cars that are not red. **b** The red electric cars. **c** The cars that are red or electric (or both). **d** There is no red electric car in the car park.

5 For two sets P and Q: $n(\mathscr{E}) = 40$, $n(P) = 22$, $n(Q) = 19$ and $n(P \cap Q) = 7$. Find $n(P \cup Q)$ and $n\big((P \cup Q)'\big)$. [3]

> Adding $n(P)$ and $n(Q)$ counts the overlap twice: $n(P \cup Q) = 22 + 19 - 7 = 34$. Outside both: $40 - 34 = 6$.

6 $\mathscr{E} = \{\text{positive integers up to } 20\}$, $M = \{\text{multiples of } 4\}$, $E = \{\text{even numbers}\}$. [4]

- **a** List M and explain why $M \subset E$.
- **b** Find $n(E \cap M')$ and describe this set.

> **a** $M = \{4, 8, 12, 16, 20\}$. Every multiple of 4 is even, so every member of M is in E: M is a subset of E.
>
> **b** E has 10 members; remove the 5 multiples of 4: $n(E \cap M') = 5$ — the even numbers that are not multiples of 4: $\{2, 6, 10, 14, 18\}$.

Total: 26 marks

Check: $\cap$ = "and" (overlap), $\cup$ = "or" (everything inside), $'$ = "not" · don't double-count the overlap · the number outside the circles is part of $\mathscr{E}$ too.